PULSEIQ ANALYTICS

PHASE 2 REPORT | PEOPLE AND TEAMS | DATA ANALYTICS TRACK

Participant	Arpita Madpalli	**Phase**	Phase 2 — Tasks 6–10
Repository	github.com/ArpitaM13/pulseiq	**Git tag**	phase-2 (verify/create before submission)
Reporting period	April 2025 – September 2026	**Dataset**	Fictional training data

Project summary

Phase 2 extends PulseIQ Analytics with attendance and leave trends, attrition analysis, team productivity, payroll reconciliation, and budget-versus-actual reporting. Streamlit dashboards use governed DuckDB analytics views, reporting filters, visual summaries, and small-group privacy thresholds. Actual employee exits are kept separate from predicted attrition risk.

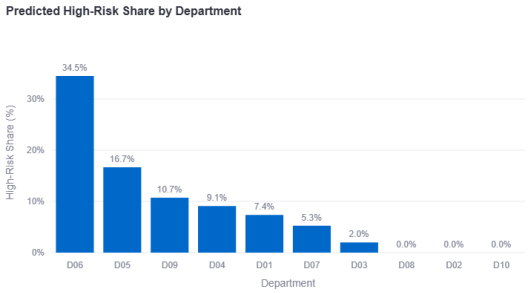
Phase 2 task status and evidence

Task	Implementation / evidence	Status
6. Attendance & Leave	Attendance and leave dashboard with reporting filters and trend visualizations. Interpret attendance changes with the working-day denominator and calendar context.	Completed
7. Attrition Analytics	Actual exits by month, department, tenure and exit type; governed M04–M06 trends; department-level predicted-risk chart and summary. Latest risk snapshot: 30 Sep 2026.	Visually verified
8. Team Productivity	Team/sprint overview and governed M22–M25 metrics: task completion, on-time delivery, sprint velocity and cycle time. Minimum group-size rule applied.	Completed
9. Payroll Reconciliation	18 of 18 months reconciled; 0 exceptions in the reported validation.	Validated
10. Budget vs Actual	Department-month budget and actual payroll compared; variance and variance percentage calculated; largest overspend and savings reviewed.	Validated

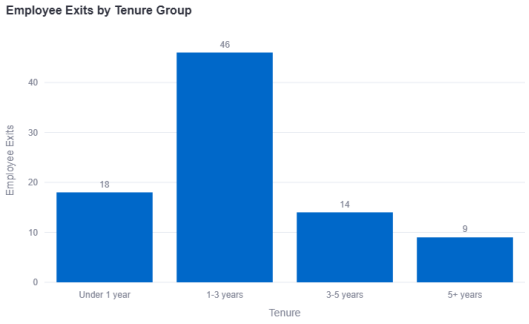
Evidence note: Task 7 screenshots are included on page 2. Add separate Task 6 and Task 8 screenshots if required by the reviewer. The final automated test result must be recorded after rerunning the local test suite.

DASHBOARD EVIDENCE AND FINDINGS

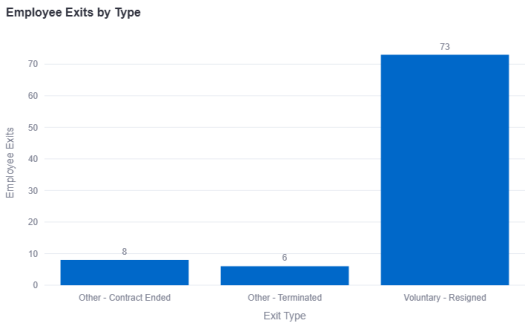
Task 7 — Attrition Analytics | screenshots from the running dashboard



Predicted high-risk share by department



Employee exits by tenure group



Employee exits by type

Key findings

Observed exits: 87 unique exits; 73 were resignations (83.9%) and 14 were other exits.

Tenure at exit: the 1–3 years group accounted for 46 exits (52.9% of observed exits). These are exit counts, not tenure-specific attrition rates.

Predicted risk: latest snapshot 30 September 2026. D06 had the highest displayed high-risk share (34.48%), followed by D05 (16.67%). Predictions are not confirmed departures.

Privacy: actual-exit groups below five and predicted-risk groups below ten are suppressed. Individual names and person-level risk scores are not displayed. Department risk is a model estimate, not a confirmed departure rate.

VALIDATION, LEARNING AND NEXT STEPS

Payroll and budget results

Measure	Reported result	Interpretation
Payroll reconciliation	18/18 months reconciled; 0 exceptions	No monthly exceptions in the reported validation.
Budget total	■853,130,000	Aggregated department-month payroll budget.
Actual payroll	■852,780,406.01	Aggregated department-month actual payroll.
Variance (actual – budget)	–■349,593.99 (–0.04%)	Actual was slightly below budget using this sign convention.
Largest overspend	D03: ■992,768.20	Review drivers and confirm whether budget assumptions remained valid.
Largest savings	D01: ■2,054,508.87	Check for efficiency, vacancies, timing or changed staffing plans.

Problems solved and governance

For attrition, employee history is ranked to select the latest record before filtering exits to the reporting period. The dashboard separates observed exits from predicted risk, applies group-size suppression, and labels exit counts separately from attrition rates. For budget analysis, variance is actual minus budget; a negative total variance means actual payroll was below budget. Known data-quality findings should remain visible for audit rather than being silently changed.

What I learned

I learned to compare attendance, attrition, delivery and payroll measures using consistent periods and governed metric views. I practiced distinguishing counts from rates, actual outcomes from predictions, and aggregate findings from individual-level data. I also learned to reconcile payroll over time and explain budget variance using a clear sign convention. Privacy thresholds and independent checks help make dashboard results more responsible and reproducible.

Submission checklist

■ Rerun the final automated tests after the last code changes and record the exact result.

■ Verify the local repository status and create/push the **phase-2** Git tag if it is not already present.

■ Confirm the GitHub repository link opens and is clickable in this PDF.

■ Add Task 6 and Task 8 screenshots if required; retain the Task 7 visual evidence.

■ Keep passwords, API keys and secrets out of GitHub; confirm this PDF is under 10 MB.

Repository: github.com/ArpitaM13/pulseiq
Filename: DataAnalytics_Arpita_Madpalli_Phase2_Report.pdf
Notice: PulseIQ Analytics uses fictional training data; findings are for training and review, not production decisions.